==========

==========

CHAPTER 4 — TCR, MAIR, AND MEMORY ATTRIBUTE CONTROL

==========

==========

A maximum-detail, RTOS/QNX-style technical breakdown of how ARMv8 controls address size, granule size, shareability, cacheability, and memory attributes through TCR and MAIR.

****

****

4.1 — Introduction to TCR and MAIR

****

****

Two registers define how the MMU interprets addresses and memory attributes:

- **TCR_EL1** — controls Stage-1 translation
- **TCR_EL2** — controls Stage-2 translation
- **MAIR_EL1** — defines Stage-1 memory types

- **MAIR_EL2** — defines Stage-2 memory types

These registers determine:

- VA size
- granule size
- page table depth
- shareability
- cacheability
- memory type selection
- IPA/PA interpretation

They are foundational to correct MMU behavior.

****

****

4.2 — TCR_EL1: Translation Control Register (Stage-1)

****

****

TCR_EL1 controls how EL0/EL1 virtual addresses are interpreted.

Major fields:

Field	Purpose
T0SZ	Size of TTBR0 VA region
T1SZ	Size of TTBR1 VA region
TG0	Granule size for TTBR0
TG1	Granule size for TTBR1

SH0	Shareability for TTBR0
SH1	Shareability for TTBR1
IRGN0	Inner cacheability for TTBR0
ORGN0	Outer cacheability for TTBR0
IRGN1	Inner cacheability for TTBR1
ORGN1	Outer cacheability for TTBR1
IPS	Intermediate Physical Address size

****

****

4.3 — TCR_EL1.T0SZ and TCR_EL1.T1SZ (VA Size Control)

****

****

These fields define the size of the virtual address space for TTBR0 and TTBR1.

Formula:

```
VA_bits = 64 - TnSZ
```

Example

If T0SZ = 16:

```
VA_bits = 64 - 16 = 48 bits
```

Teaching Note

This determines how many VA bits are valid and how many must be sign-extended.

****

****

4.4 — Granule Size (TG0/TG1)

****

****

Granule size determines:

- page size
- page table depth
- VA slicing
- descriptor interpretation

Supported granules:

- **4 KB** (TGx = 00)
- **16 KB** (TGx = 01)
- **64 KB** (TGx = 10)

VA slicing for 4 KB granule:

```
VA[47:39] → L0 (optional)
VA[38:30] → L1
VA[29:21] → L2
VA[20:12] → L3
VA[11:0]  → page offset
```

Implementation Note

Granule size must match page table layout in memory.

****

****

4.5 — Shareability (SH0/SH1)

****

****

Shareability controls cache coherence across cores.

Values:

- **00 = Non-shareable**
- **10 = Outer Shareable**
- **11 = Inner Shareable**

Teaching Note

Inner Shareable is required for SMP systems.

****

****

4.6 — Cacheability (IRGN/ORGN)

****

****

Defines inner/outer cacheability for page table walks.

Values:

- **00 = Non-cacheable**
- **01 = Write-back cacheable**
- **10 = Write-through cacheable**
- **11 = Write-back cacheable (read-allocate)**

Implementation Note

Page tables should be **Write-Back Inner Shareable** for performance.

****

****

4.7 — TCR_EL1.IPS (IPA Size)

****

****

Defines the size of the Intermediate Physical Address (IPA):

IPS	IPA Size
000	32 bits
001	36 bits
010	40 bits
011	42 bits
100	44 bits
101	48 bits

Teaching Note

IPA size must match hardware PA range.

****

****

4.8 — TCR_EL2: Translation Control Register (Stage-2)

****

****

TCR_EL2 controls how IPA is interpreted during Stage-2 translation.

Fields mirror TCR_EL1:

- T0SZ → IPA size
- TG0 → granule size
- SH0 → shareability
- IRGN0/ORGN0 → cacheability
- PS → physical address size

Implementation Note

Hypervisor must configure TCR_EL2 before enabling Stage-2 MMU.

****

****

4.9 — MAIR_EL1: Memory Attribute Indirection Register

****

****

MAIR_EL1 defines **8 memory types** for Stage-1.

Each entry is 8 bits:

```
MAIR_EL1 = [Attr7][Attr6]...[Attr0]
```

Each AttrX defines:

- device vs normal
- cacheability
- ordering rules

****

****

4.10 — MAIR_EL2: Stage-2 Memory Attributes

****

****

MAIR_EL2 defines memory types for Stage-2.

Hypervisor uses MAIR_EL2 to override guest memory types.

Example

Guest marks MMIO as Normal WB. Hypervisor forces Device-nGnRnE.

****

****

4.11 — Memory Attribute Encoding

****

****

Normal Memory

Type	Encoding
Write-Back	0xFF
Write-Through	0xBB
Non-cacheable	0x44

Device Memory

Type	Encoding
Device-nGnRnE	0x00
Device-nGnRE	0x04
Device-nGRE	0x08

Teaching Note

Device-nGnRnE is the safest device type.

****

****

4.12 — Memory Attribute Selection (AttrIndx)

****

****

Page table descriptors contain:

`AttrIndx[2:0]`

This selects one of the 8 MAIR entries.

Example:

`AttrIndx = 3 → MAIR_EL1[3]`

Implementation Note

Incorrect AttrIndx causes cache corruption or device misbehavior.

****

****

4.13 — Stage-1 vs Stage-2 Attribute Merging

****

****

Final memory type = Stage-2 overrides Stage-1.

Examples:

- Guest: Normal WB
- Hypervisor: Device-nGnRnE → Final: Device-nGnRnE
- Guest: Normal NC
- Hypervisor: Normal WB → Final: Normal WB

Teaching Note

Hypervisor always wins.

****

****

4.14 — TCR and MAIR Interaction

****

****

TCR defines:

- granule
- VA size
- shareability
- cacheability

MAIR defines:

- memory type
- device vs normal
- cacheability rules

Together they determine:

- page table walk behavior
- memory access behavior
- cache behavior
- ordering rules

****

****

4.15 — Example: Configuring TCR_EL1 for Linux/QNX

****

****

Typical configuration:

```
T0SZ = 16        → 48-bit VA
TG0  = 00        → 4 KB granule
SH0  = 11        → Inner Shareable
IRGN0= 01        → Write-Back
ORGN0= 01        → Write-Back
IPS  = 010       → 40-bit IPA
```

MAIR_EL1:

```
Attr0 = 0xFF → Normal WB
Attr1 = 0x44 → Normal NC
Attr2 = 0x00 → Device-nGnRnE
```

****

****

4.16 — Example: Configuring TCR_EL2 for Hypervisor

****

****

Typical hypervisor configuration:

```
T0SZ = 24        → 40-bit IPA
TG0  = 00        → 4 KB granule
SH0  = 11        → Inner Shareable
IRGN0= 01        → Write-Back
ORGN0= 01        → Write-Back
PS   = 010       → 40-bit PA
```

MAIR_EL2:

```
Attr0 = 0xFF → Normal WB
Attr1 = 0x00 → Device-nGnRnE
```

****

****

4.17 — TCR Fault Conditions

****

****

Faults occur if:

- invalid granule size selected
- unsupported PA range selected
- TnSZ produces invalid VA size
- shareability conflicts with memory type

Fault type: **Translation Fault** or **Address Size Fault**.

****

****

4.18 — MAIR Fault Conditions

****

****

Faults occur if:

- AttrIndx references uninitialized MAIR entry
- device memory used with cacheable attributes

- normal memory used with device attributes

Fault type: **Permission Fault** or **Alignment Fault**.

****

****

4.19 — Chapter Summary

****

****

This chapter covered:

- TCR_EL1 and TCR_EL2
- granule size
- VA size
- shareability
- cacheability
- IPA/PA size
- MAIR_EL1 and MAIR_EL2
- memory attribute encoding
- Stage-1 vs Stage-2 attribute merging
- full configuration examples
- fault conditions